Ambery (Yu-Jung) Liang

La Jolla, CA | amliang@ucsd.edu | Medium | Linkedin | Github

Education

University of California San Diego, MS in Electrical & Computer Engineering Sep. 2025 – Jun. 2027
(Specialization: Machine Learning and Data Science)

National Chung Hsing University, BBA in Information Management Sep. 2021 – Jun. 2025

Research Experience

Debate-Time Defense Against Insider Collusion in Medical Multi-Agent Systems Jun. 2026 – present
Independent research (targeting SaTML 2027)

- Designed a debate-time collusion probe — publishing per-agent reliability scores and randomly depressing one agent's score each round — that tests for mutual insider cover-up via a difference-in-differences (DiD) rating shift, after five prior single-agent behavioral and verifiable-claim detectors failed to generalize across model scales (one detector's separation inverted at 14B: honest 49.1% vs. insider 26.7%).
- On a 5-agent medical debate panel (MedLA, MDAgents, Qwen2.5-Coder-7B): teammate-unaware insiders show no cover-up signal (DiD +0.18 to +0.20, 95% CI spans zero under both fed-back and shadow judge configurations) while teammate-aware insiders show a significant cover-up signal (DiD +0.50 to +0.81, CI excludes zero) that replicates across three independent runs.
- Paired the collusion probe with a rotating verifiable-claim judge in fed-back vs. shadow (announced-but-not-relayed) configurations: shadow judge + teammate-aware insiders yields both the strongest cover-up signal (DiD +0.81) and the best debate accuracy (47%), while pairing against a no-probe rotating-judge baseline (36%) confirms the added score channel does not degrade outcomes (12 vs. 8 discordant questions, statistically null).

AI Engineer, Far Eastern Memorial Hospital — Remote, Taiwan Jun. 2025 – May 2026
Advisor: Dr. Fang-Ming Hung

- Built a knowledge-graph multi-agent clinical decision-support system for cardiology, orchestrated with LangGraph, on de-identified electronic health records, framed to reduce clinician burnout and cognitive load; a Medical-NER module parses free-text admission notes into a longitudinal Neo4j knowledge graph, feeding a supervised fine-tuned ICD-10 prediction agent plus treatment and medication recommendation agents with drug-interaction checks.
- Designed a dual-source fusion step grounding LLM generation in RAG-retrieved analogous cases to hedge hallucination, and an active-learning loop that writes physician edits back to the graph as "correction guidelines"; designed a pre/post well-being evaluation for validation.

Undergraduate Researcher, National Science Council — NCHU, Taiwan Sep. 2023 – Mar. 2025
Advisor: Prof. Meng-Hsiun Tsai

- Built an end-to-end hybrid semantic-segmentation and object-detection pipeline (U-Net + YOLOv8) for breast-ultrasound lesions, reaching 95% precision and 92% recall on 3,000+ labeled scans, with prediction-overlay error analysis of low-contrast failure cases.

Publications

To Diffuse or Not to Diffuse: Matching Graph Retrieval to Memory Structure in LLM Agents — accepted, Lifelong Agent Workshop @ COLM 2026 COLM 2026

- Built a 5W (who/what/when/where/why) extraction pipeline — LLM tagging with a canonicalization layer — feeding a graph-traversal retrieval method, with a vocabulary-normalization upgrade (generic-word filtering + TF-IDF-weighted overlap) that recovered +62.6% relative recall@1 within the graph retriever.
- Isolated retrieval algorithm from extraction quality via a controlled harness on LongMemEval/ATANT holding notes fixed across retrievers (dense recall@5 82.5 vs. graph-traversal 41.8–53.7), with paired-bootstrap testing (10k resamples, $p \leq 0.001$) confirming the gap is algorithm-bound, not extraction-bound; results pin down when graph-structured memory's extra extraction cost earns its keep versus when it merely adds diffusion overhead over flat dense retrieval.

Selected Projects

Stale-but-Valid — Real-Time Speech-Agent Coordinator Jul. 2026 – present

- Reference coordinator that hides slow-reasoner (LLM) latency behind stale-but-valid reads of a typed, versioned belief state, with explicit freshness classes (fresh / stale-valid / expired) and a four-token repair vocabulary (revise / retract / extend / defer) for mid-utterance correction, each mapped to a named recovery action.

Technical Skills

ML / DL: PyTorch, Transformers/BEHRT, XGBoost, Scikit-learn, multi-agent LLMs, RAG, LangGraph, RL
Data & Infra: Pandas, SQL, Neo4j, NetworkX, Docker, Git, Linux, LM Studio / OpenAI-compatible serving
Languages: Python, TypeScript, SQL, Bash · **Focus**: adversarial robustness, medical AI, agent systems